

Internal review draft

Type-1, $N = 3$ Landau–Zener Double Cover, Reduced ODE, Gluing, and Selector

Assumed background. The reader already knows the exact Type-1 spectral parametrization, the augmented solver, and the gauged adiabatic-basis evolution summarized in `LZ_summary`. This primer begins *after* that point.

Goal of this draft. Organize the post-summary story into one technically motivated backbone:

1. start from the directly computable pair spaces (u, Δ) , (u, S) , and (Δ, S) ,
2. build the common cover and the spinor double cover,
3. define the meromorphic half-form amplitude package adapted to that geometry,
4. derive the compact quotient-ring evolution

$$\partial_u[R] = \mathbf{i}[(m - E_x p)s_u R] \quad \text{in } \mathcal{E}_u,$$

5. identify the same geometry operator-theoretically via the commuting triple (H_x, Y, R) ,
6. state and prove the canonical gluing theorem,
7. end with the selector criterion and the data still needed to recover transition probabilities.

Notation alert. The document uses $[R] \in \mathcal{E}_u$ for the reduced amplitude class and later R for the transport multiplication operator. The font difference is intentional and important.

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1 Standing notation and scope

We work throughout in the Type-1, $N = 3$ setting. The exact spectral parametrization gives rational functions

$$u(\lambda) = \frac{n(\lambda)}{p(\lambda)}, \quad E(\lambda) = \frac{m(\lambda)}{p(\lambda)}, \quad q_u(\lambda) := u p(\lambda) - n(\lambda), \quad (1.1)$$

with three roots $\lambda_0(u), \lambda_1(u), \lambda_2(u)$ of q_u at each regular base point u .

1.1 Coefficient conventions for p , n , m , and the chosen family member

For $N = 3$ we normalize

$$\begin{aligned} p(\lambda) &= \lambda^3 + p_2 \lambda^2 + p_1 \lambda + p_0, & p_2 &= -(\epsilon_0 + \epsilon_1 + \epsilon_2), & p_1 &= \epsilon_0 \epsilon_1 + \epsilon_0 \epsilon_2 + \epsilon_1 \epsilon_2, & p_0 &= -\epsilon_0 \epsilon_1 \epsilon_2, \\ n(\lambda) &= n_2 \lambda^2 + n_1 \lambda + n_0, & n_2 &= \sum_{a=0}^2 \gamma_a^2, & n_1 &= -\sum_{a=0}^2 \gamma_a^2 \sum_{b \neq a} \epsilon_b, & n_0 &= \sum_{a=0}^2 \gamma_a^2 \prod_{b \neq a} \epsilon_b, \\ m(\lambda) &= m_2 \lambda^2 + m_1 \lambda + m_0, & m_2 &= \sum_{a=0}^2 a_a \gamma_a^2, & m_1 &= -\sum_{a=0}^2 a_a \gamma_a^2 \sum_{b \neq a} \epsilon_b, & m_0 &= \sum_{a=0}^2 a_a \gamma_a^2 \prod_{b \neq a} \epsilon_b. \end{aligned} \quad (1.2)$$

The chosen commuting-family member H is equivalently encoded by the unique quadratic interpolation polynomial

$$f_H(\lambda) = \alpha_H \lambda^2 + \beta_H \lambda + \gamma_H, \quad f_H(\epsilon_a) = a_a \quad (a = 0, 1, 2). \quad (1.3)$$

Then

$$m(\lambda) = f_H(\lambda) n(\lambda) - p(\lambda) \ell_H(\lambda), \quad \ell_H(\lambda) = \alpha_H \Gamma_{\text{tot}} \lambda + \sum_{a=0}^2 \gamma_a^2 (\alpha_H \epsilon_a + \beta_H), \quad (1.4)$$

where $\Gamma_{\text{tot}} := \sum_{a=0}^2 \gamma_a^2 = n_2$.

From `LZ_summary` we take as starting point the gauged adiabatic amplitudes $f_i(u)$ satisfying

$$f'_i(u) = -\sum_{j \neq i} S_{ij}(u) f_j(u) + i(E_i(u) - E_x(u)) f_i(u), \quad i = 0, 1, 2. \quad (1.5)$$

Here f_i are coefficients in the moving adiabatic frame, the diagonal part $E_i - E_x$ carries phase accumulation, and the antisymmetric matrix S_{ij} is the transport connection of that frame. The only additional baseline fact used below is that S_{ij} is smooth through $u = 0$.

The technical progression developed in this primer is

$$(1.5) \implies \partial_u R(\lambda_i(u); u) = i(E_i - E_x) R(\lambda_i(u); u) \implies \partial_u [R] = i[(m - E_x p) s_u R].$$

The first step passes from the moving adiabatic frame to a reduced spectral carrier; the second

repackages the three sheetwise scalar equations into one equation in the quotient-ring bundle.

Guiding question. Starting from the three directly computable pair data sets

$$(u, \Delta_{ij}), \quad (u, S_{ij}), \quad (\Delta_{ij}, S_{ij}),$$

what is the smallest global object on which the exact-WKB phase data, the transport data, and the reduced Type-1 evolution all become part of one story?

2 Formal I. From pair spaces to the common cover and the spinor double cover

2.1 The directly computable pair spaces

Let $U \subset \mathbb{P}_u^1$ be simply connected and avoid ordinary turning points and transport ramification points. For each spectator index $i \in \{0, 1, 2\}$, let

$$\Delta_i(u) = E_j(u) - E_k(u), \quad S_i(u) = S_{jk}(u), \quad \{i, j, k\} = \{0, 1, 2\}.$$

These are the local pair quantities attached to the complementary pair (j, k) .

Definition 2.1 (Pair spaces). The three pair spaces are the normalizations of the closures of the local graphs

$$\Sigma_{u,\Delta} := \overline{\{(u, \Delta_i(u))\}}, \tag{2.1}$$

$$\Sigma_{u,S} := \overline{\{(u, S_i(u))\}}, \tag{2.2}$$

$$\Sigma_{\Delta,S} := \overline{\{(\Delta_i(u), S_i(u))\}}, \tag{2.3}$$

as the spectator label i ranges over $\{0, 1, 2\}$.

These are precisely the spaces that polynomial elimination or direct numerical continuation sees first.

2.2 Sign-invariant coordinates

The pair variables carry sign ambiguities: Δ_i changes sign when the complementary pair is exchanged, while S_i changes sign with the transport square root. The combinations

$$y_i := \Delta_i^2, \quad r_i := \kappa_S \frac{S_i^{-2}}{\Delta_i^4} \tag{2.4}$$

are invariant under both sign changes.

Here $\kappa_S \in \mathbb{C}^\times$ is the normalization constant that converts the sheetwise transport invariant into the rational commuting-family coordinate r . Proposition 2.2 will show that the natural $N = 3$ choice is

$$\kappa_S = -\text{Res}_\lambda(p, n) = \gamma_0^2 \gamma_1^2 \gamma_2^2 \prod_{a < b} (\epsilon_a - \epsilon_b)^2,$$

so in the real ordered Type-1 family it is in fact positive. The sign that matters geometrically is the sign of the spinor variable ρ , not a hidden sign of κ_S .

For $N = 3$, the squared gaps are not independent. If e_0, e_1, e_2 are the three adiabatic energies, define

$$A(u) := e_0^2 + e_1^2 + e_2^2 - e_0 e_1 - e_0 e_2 - e_1 e_2, \quad D(u) := \prod_{a < b} (e_a - e_b)^2. \quad (2.5)$$

Then each y_i satisfies the universal cubic

$$\chi_Y(u, y) := y^3 - 2A(u)y^2 + A(u)^2 y - D(u) = 0. \quad (2.6)$$

Equivalently, each gap Δ satisfies the sextic

$$\Delta^6 - 2A(u)\Delta^4 + A(u)^2\Delta^2 - D(u) = 0. \quad (2.7)$$

Proposition 2.2 (Coefficient-level formulas for Q_4 , L_H , W_4 , and κ_S). *Fix a spectator coordinate ξ and define the complementary quadratic*

$$g_\xi(z) = \frac{n(z)p(\xi) - n(\xi)p(z)}{z - \xi} = A_\xi z^2 + B_\xi z + C_\xi, \quad (2.8)$$

with

$$A_\xi = -n(\xi), \quad B_\xi = n_2 p_1 \xi + n_2 p_0 - n_1 \xi^2 - n_1 p_2 \xi - n_0 \xi - n_0 p_2, \quad C_\xi = n_2 p_0 \xi + n_1 p_0 - n_0 \xi^2 - n_0 p_2 \xi - n_0 p_1. \quad (2.9)$$

Let $z_\pm(\xi)$ be the two roots of g_ξ . Define

$$Q_4(\xi) := \text{Disc}_z(g_\xi) = B_\xi^2 - 4A_\xi C_\xi = B_\xi^2 + 4n(\xi)C_\xi. \quad (2.10)$$

Equivalently,

$$Q_4(\lambda) = q_4 \lambda^4 + q_3 \lambda^3 + q_2 \lambda^2 + q_1 \lambda + q_0, \quad (2.11)$$

where

$$\begin{aligned}
q_4 &= n_1^2 - 4n_0n_2, \\
q_3 &= 2(-n_0n_1 - 2n_0n_2p_2 + n_1^2p_2 - n_1n_2p_1 + 2n_2^2p_0), \\
q_2 &= -3n_0^2 - 6n_0n_2p_1 + n_1^2p_2^2 + 6n_1n_2p_0 - 2n_1n_2p_1p_2 + n_2^2p_1^2, \\
q_1 &= 2(-n_0^2p_2 - 2n_0n_1p_1 + n_0n_1p_2^2 + n_0n_2p_0 - n_0n_2p_1p_2 + 2n_1^2p_0 - n_1n_2p_0p_2 + n_2^2p_0p_1), \\
q_0 &= -4n_0^2p_1 + n_0^2p_2^2 + 4n_0n_1p_0 - 2n_0n_2p_0p_2 + n_2^2p_0^2.
\end{aligned}$$

For the chosen family member H , define the linear factor

$$L_H(\lambda) := \beta_H + \alpha_H(\epsilon_0 + \epsilon_1 + \epsilon_2 - \lambda). \quad (2.12)$$

Define also the quartic transport polynomial

$$W_4(\lambda) := n(\lambda)p'(\lambda) - n'(\lambda)p(\lambda) = p(\lambda)^2\Sigma^{(2)}(\lambda), \quad (2.13)$$

so explicitly

$$W_4(\lambda) = n_2\lambda^4 + 2n_1\lambda^3 + (3n_0 + n_1p_2 - n_2p_1)\lambda^2 + 2(n_0p_2 - n_2p_0)\lambda + (n_0p_1 - n_1p_0). \quad (2.14)$$

Finally set

$$\kappa_S := -\text{Res}_\lambda(p, n) = \gamma_0^2\gamma_1^2\gamma_2^2 \prod_{a < b} (\epsilon_a - \epsilon_b)^2. \quad (2.15)$$

Then the sign-invariant phase and transport coordinates on the common sheet are

$$y(\lambda) = \frac{Q_4(\lambda)L_H(\lambda)^2}{p(\lambda)^2}, \quad S(\lambda)^{-2} = \frac{Q_4(\lambda)^2W_4(\lambda)}{\kappa_S p(\lambda)^4}, \quad r_H(\lambda) = \frac{W_4(\lambda)}{L_H(\lambda)^4}. \quad (2.16)$$

Proof. Write the complementary energy gap as

$$\Delta(\xi) = E(z_+) - E(z_-).$$

Using (1.4), the difference formula from the $N = 3$ gap reduction becomes

$$\Delta(\xi) = (z_+ - z_-) \left(u(\xi)(\alpha_H(z_+ + z_-) + \beta_H) - \alpha_H \Gamma_{\text{tot}} \right).$$

Now $z_+ + z_- = -B_\xi/A_\xi = B_\xi/n(\xi)$ and $u(\xi) = n(\xi)/p(\xi)$, so the bracket collapses to

$$\frac{\alpha_H B_\xi + \beta_H n(\xi) - \alpha_H \Gamma_{\text{tot}} p(\xi)}{p(\xi)} = \frac{n(\xi)L_H(\xi)}{p(\xi)}.$$

Since $(z_+ - z_-)^2 = \text{Disc}_z(g_\xi)/A_\xi^2 = Q_4(\xi)/n(\xi)^2$, this gives the first identity in (2.16).

For transport, one has

$$S(\xi)^{-2} = (z_+ - z_-)^2 \Sigma^{(2)}(z_+) \Sigma^{(2)}(z_-), \quad \Sigma^{(2)}(z) = \frac{W_4(z)}{p(z)^2}.$$

Hence

$$S(\xi)^{-2} = \text{Disc}_z(g_\xi) \frac{\text{Res}_z(g_\xi, W_4)}{\text{Res}_z(g_\xi, p)^2}.$$

A direct coefficient computation gives the two exact factorization identities

$$\text{Res}_z(g_\xi, p) = \kappa_S p(\xi)^2, \quad \text{Res}_z(g_\xi, W_4) = \kappa_S W_4(\xi) Q_4(\xi). \quad (2.17)$$

Substituting (2.17) yields the second identity in (2.16); dividing by $y(\lambda)^2$ gives the third. $\square$

Remark 2.3 (What κ_S really does). For the real ordered Type-1 family, (2.15) shows that κ_S is strictly positive, so it is not carrying a hidden physical sign choice. Its job is to absorb the family-invariant resultant in (2.17) so that $r_H = W_4/L_H^4$ becomes the natural rational transport coordinate. The sign that genuinely matters is the sign of the spinor square root ρ ; changing $\rho \mapsto -\rho$ is the deck involution of the transport lift. More generally, multiplying κ_S by any nonzero constant only rescales r and ρ and leaves the ramification set, the common cover, and the selector geometry unchanged.

Definition 2.4 (Early transport ramification locus). The transport ramification divisor on the common-sheet sphere is cut out by $W_4(\lambda) = 0$. Its image in the base,

$$\mathfrak{C}_R := u(\{\lambda \in \mathbb{P}^1 : W_4(\lambda) = 0\}), \quad (2.18)$$

is the transport discriminant locus. Equivalently, these are the base points where the degree-three map $u = n/p$ has a critical point and hence where the normalized transport curve ramifies over $\mathbb{P}_u^1$. Section 6 will restate the same set as the discriminant of the cubic spectral polynomial $\chi_R(u, r)$.

Definition 2.5 (Phase and transport curves). The *phase curve* is

$$\Sigma_Y := \{(u, y) \in \mathbb{P}_u^1 \times \mathbb{C} : \chi_Y(u, y) = 0\}. \quad (2.19)$$

The *transport curve* Σ_R is the normalization of the closure of the local graphs $(u, r_i(u))$.

On any regular simply connected domain one may equivalently package the transport branches by the local cubic

$$\chi_R^{\text{loc}}(u, r) := \prod_{i=0}^2 (r - r_i(u)).$$

Its coefficients are symmetric in the local branches, hence descend meromorphically to the base. Thus the transport curve may be written formally as

$$\Sigma_R = \{(u, r) \in \mathbb{P}_u^1 \times \mathbb{C} : \chi_R^{\text{loc}}(u, r) = 0\},$$

and Section 5 will identify this same curve operator-theoretically as the spectral curve of R .

To obtain the honest (u, S^{-2}) curve one must eliminate using the *same* local sheet that selects the complementary pair.

Proposition 2.6 (The compatible (u, S^{-2}) polynomial). *Set $\sigma := S^{-2}$. Fix a local sheet parameter $\xi = \lambda_i(u)$, and define the quadratic*

$$g_\xi(z) := \frac{n(z)p(\xi) - n(\xi)p(z)}{z - \xi}. \quad (2.20)$$

Its two roots $z_\pm(\xi)$ are the other preimages of $u(\xi)$, hence the complementary pair to the sheet ξ . Since

$$\sigma(\xi) = (z_+ - z_-)^2 \Sigma^{(2)}(z_+) \Sigma^{(2)}(z_-)$$

and $\Sigma^{(2)}(z) = W_4(z)/p(z)^2$, one gets the sheetwise rational expression

$$\sigma(\xi) = \text{Disc}_z(g_\xi) \frac{\text{Res}_z(g_\xi, W_4)}{\text{Res}_z(g_\xi, p)^2} = \frac{N_{12}(\xi)}{D_{12}(\xi)}. \quad (2.21)$$

Here N_{12} and D_{12} are degree-12 polynomials in ξ . Eliminating ξ against the cubic equation $q_u(\xi) = 0$ gives the compatible branch polynomial

$$\chi_{S^{-2}}(u, \sigma) := \text{Res}_\xi(q_u(\xi), D_{12}(\xi)\sigma - N_{12}(\xi)) = 0. \quad (2.22)$$

After monic normalization,

$$\chi_{S^{-2}}(u, \sigma) = \sigma^3 + c_1(u)\sigma^2 + c_2(u)\sigma + c_3(u), \quad \deg_u c_k \leq 4k. \quad (2.23)$$

So the (u, S^{-2}) space is cut out by a polynomial that is cubic in σ and twelfth order in the base variable u .

Proof. The coefficients of g_ξ are quadratic in ξ . Hence $\text{Disc}_z(g_\xi)$ is quartic in ξ , $\text{Res}_z(g_\xi, W_4)$ has degree 8 in ξ , and $\text{Res}_z(g_\xi, p)$ has degree 6 in ξ , which proves (2.21). The resultant (2.22) is therefore cubic in σ . Since $q_u(\xi)$ is linear in u and cubic in ξ , while $D_{12}(\xi)\sigma - N_{12}(\xi)$ is linear in σ and degree 12 in ξ , the u -degree is 12, and the coefficient bounds in (2.23) follow. $\square$

This is the correct analogue of the gap sextic (2.7). A naive elimination using the separate phase and transport curves would allow an arbitrary y -branch and an arbitrary r -branch above the same base point. That overcounts the true (u, S^{-2}) space because it forgets compatibility. Technically this is also why r is the more useful transport coordinate: it belongs to a commuting family on its own, whereas S^{-2} is only recovered after one has fixed the common sheet.

2.3 The common cover and its spinor lift

Definition 2.7 (Common phase-transport cover). The common cover is the fiber product

$$\mathcal{C} := \Sigma_Y \times_{\mathbb{P}_u^1} \Sigma_R. \quad (2.24)$$

A point of $\mathcal{C}$ records one base point together with one compatible phase branch and one compatible transport branch.

The subscript $\times_{\mathbb{P}_u^1}$ means precisely that the chosen phase branch and transport branch lie above the same base point u . The order of the factors is immaterial:

$$\Sigma_Y \times_{\mathbb{P}_u^1} \Sigma_R \cong \Sigma_R \times_{\mathbb{P}_u^1} \Sigma_Y.$$

We place Σ_Y first only to match the narrative order of the construction.

On $\mathcal{C}$, the sign-invariant quantities are single-valued rational functions:

$$y = \Delta^2, \quad r = \kappa_S \frac{S^{-2}}{\Delta^4}. \quad (2.25)$$

Definition 2.8 (Spinor double cover). The spinor double cover is the two-fold cover

$$\widehat{\mathcal{C}} := \{(u, r, y, \rho) \in \mathcal{C} \times \mathbb{C} : \rho^2 = \kappa_S/r\}. \quad (2.26)$$

The lifted variable ρ is the transport square root.

Proposition 2.9 (How the double cover is built from pair data, and how pair data are recovered).

Locally on $\widehat{\mathcal{C}}$, choose a sign of $\sqrt{y}$ and write $\Delta := \sqrt{y}$. Then

$$S = \frac{\rho}{y}, \quad \rho = \Delta^2 S. \quad (2.27)$$

Consequently:

1. *the pair spaces are recovered from $\widehat{\mathcal{C}}$ by the local projections*

$$(u, r, y, \rho) \mapsto (u, \Delta), \quad (u, r, y, \rho) \mapsto (u, S), \quad (u, r, y, \rho) \mapsto (\Delta, S);$$

2. *the common cover $\mathcal{C}$ is created from the pair spaces by passing from (Δ, S) to the sign-invariant coordinates (y, r) ;*
3. *the spinor double cover is created from $\mathcal{C}$ by adjoining the transport square root ρ .*

Proof. The first identity in (2.27) follows immediately from (2.4) and (2.26):

$$\rho^2 = \frac{\kappa_S}{r} = \Delta^4 S^2.$$

Choosing the local sign of ρ fixes the sign of S , hence $S = \rho/y$. The projection statements are then tautological. Conversely, if one starts from the pair spaces, the only combinations that survive the local sign ambiguities are exactly $y = \Delta^2$ and $r = \kappa_S S^{-2}/\Delta^4$, so those are the coordinates that glue globally. Adjoining ρ is precisely the final square-root step that restores the transport sign. $\square$

2.4 Family invariance and topology of the normalized spaces

The formulas above separate the genuinely family-invariant polynomials p, n, Q_4, W_4, κ_S from the member-dependent linear factor L_H .

Proposition 2.10 (The same commuting family shares the same abstract covers). *For a fixed Type-1 commuting family, the data p, n, Q_4, W_4 , and κ_S depend only on (ϵ_a, γ_a) ; the chosen family member H enters only through L_H . Consequently the normalized abstract curves*

$$\mathcal{C}^{\text{norm}} \cong \mathbb{P}_\lambda^1, \quad \widehat{\mathcal{C}}^{\text{norm}} \cong \{(\lambda, \nu) : \nu^2 = W_4(\lambda)\},$$

and likewise the phase and mixed signed covers

$$C_Q := \{(\lambda, \mu) : \mu^2 = Q_4(\lambda)\}, \quad C_{Q,W} := \{(\lambda, \mu, \nu) : \mu^2 = Q_4(\lambda), \nu^2 = W_4(\lambda)\},$$

are common to the whole commuting family. Different members merely change the coordinate functions induced on those spaces:

$$y_H = \frac{Q_4 L_H^2}{p^2}, \quad r_H = \frac{W_4}{L_H^4}, \quad \Delta_H = \mu \frac{L_H}{p}, \quad \rho_H = \sqrt{\kappa_S} \frac{L_H^2}{\nu}.$$

In particular, members of the same commuting family use the same transport curve, the same common cover, and the same spinor double cover as abstract curves; what changes is the coordinate in which a given member sees them.

Proof. The family invariance of p and n is built into the Type-1 data: they depend only on (ϵ_a, γ_a) . Proposition 2.2 then shows that Q_4, W_4 , and κ_S are determined by p and n , while H enters only through L_H . The displayed formulas therefore describe the same abstract covers with different rational coordinate functions attached to different commuting-family members. $\square$

Proposition 2.11 (Generic genera and branch structure). *Assume throughout this subsection that Q_4 and W_4 have simple zeros and no common zero. Then:*

$$g(\Sigma_Y^{\text{norm}}) = g(\Sigma_R^{\text{norm}}) = g(\mathcal{C}) = 0, \quad g(\Sigma_{u,\Delta}) = g(\Sigma_{u,S}) = g(\widehat{\mathcal{C}}) = 1, \quad g(\Sigma_{\Delta,S}) = 5. \quad (2.28)$$

More precisely:

1. $\mathcal{C}$ is the common-sheet sphere $\mathbb{P}_\lambda^1$ with degree-three map $u = n/p$ to the base.
2. The ramification divisor of $u : \mathcal{C} \rightarrow \mathbb{P}_u^1$ is cut out by $W_4(\lambda) = 0$; it has degree four, and its image is exactly $\mathfrak{C}_R$.
3. The phase pair space $\Sigma_{u,\Delta}$ is birational to the double cover $\mu^2 = Q_4(\lambda)$, hence is generically elliptic.
4. The transport pair space $\Sigma_{u,S}$ and the spinor double cover $\hat{\mathcal{C}}$ are birational to the double cover $\nu^2 = W_4(\lambda)$, hence are also generically elliptic.
5. The mixed pair space $\Sigma_{\Delta,S}$ is birational to the biquadratic cover $\mu^2 = Q_4(\lambda)$, $\nu^2 = W_4(\lambda)$ with deck group $(\mathbb{Z}/2\mathbb{Z})^2$.

In particular,

$$b_1(\Sigma_{u,\Delta}) = b_1(\Sigma_{u,S}) = b_1(\hat{\mathcal{C}}) = 2, \quad b_1(\Sigma_{\Delta,S}) = 10.$$

If zeros collide or Q_4 and W_4 meet, the normalized genera drop; the displayed values are the generic $N = 3$ topology relevant to the selector problem.

Proof. The rational parameterization $(u, y, r) = (n/p, Q_4 L_H^2/p^2, W_4/L_H^4)$ identifies the normalizations of Σ_Y , Σ_R , and $\mathcal{C}$ with the spectator sphere $\mathbb{P}_\lambda^1$, so each has genus zero. The degree-three map $u = n/p$ ramifies exactly where $u'(\lambda) = 0$, i.e. where $W_4(\lambda) = n(\lambda)p'(\lambda) - n'(\lambda)p(\lambda)$ vanishes. Since $\deg W_4 = 4$, the total ramification is four, as required by Riemann–Hurwitz for a degree-three map from $\mathbb{P}^1$ to $\mathbb{P}^1$.

For $\Sigma_{u,\Delta}$, the only nontrivial extension is the phase square root $\mu^2 = Q_4(\lambda)$. Because Q_4 is a quartic with four simple finite branch points and no branch at infinity, Riemann–Hurwitz gives genus one. The same argument applies to $\Sigma_{u,S}$ and $\hat{\mathcal{C}}$, whose nontrivial square root is $\nu^2 = W_4(\lambda)$.

For $\Sigma_{\Delta,S}$, the function field is the compositum $\mathbb{C}(\lambda)(\sqrt{Q_4}, \sqrt{W_4})$. Its three index-two quotients are the double covers defined by Q_4 , by W_4 , and by $Q_4 W_4$. Under the genericity hypothesis, these have genera 1, 1, and 3 respectively; for a biquadratic cover of $\mathbb{P}^1$, the total genus is the sum of those three quotient genera. Hence $g(\Sigma_{\Delta,S}) = 1 + 1 + 3 = 5$. The Betti-number statements are $b_1 = 2g$. $\square$

Companion intuition.

The double cover is not a mysterious extra gadget. It appears because the pair data are over-specified locally and under-specified globally.

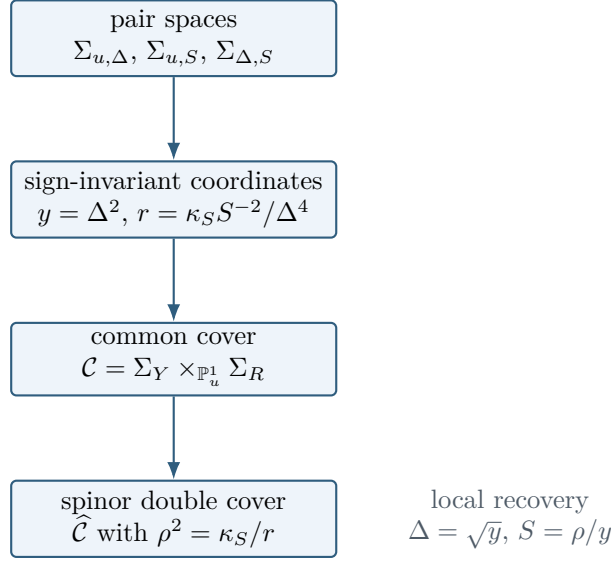


Figure 1: The backbone of the construction. The directly computable pair spaces are not discarded; they are reorganized through the sign-invariant coordinates y and r , then lifted by the transport square root.

The local computations naturally produce (u, Δ) , (u, S) , and (Δ, S) . But each of those descriptions still remembers signs that are not intrinsically global. The combinations $y = \Delta^2$ and $r = \kappa_S S^{-2}/\Delta^4$ are what survive continuation. They are the coordinates that glue. The reason to privilege r is technical, not cosmetic: the quartic W_4 and its critical-value image $\mathfrak{C}_R$ are common to the whole commuting family, whereas the visible coordinates S and r_H depend on the chosen member only through the harmless linear factor L_H . In other words, r_H sees the family-invariant transport branching before a transport square root has been chosen; S reappears only after the spinor lift.

Topologically, the picture is also cleaner than it first appears. The common cover $\mathcal{C}$ is just the spectator sphere $\mathbb{P}_\lambda^1$. Above it sit one phase double cover branched at the four zeros of Q_4 , one transport double cover branched at the four zeros of W_4 , and their mixed biquadratic cover. Generically that means one rational common cover, two elliptic one-square-root covers, and one genus-five fully signed pair space.

So the correct mental picture is:

pair spaces first $\longrightarrow$ family-invariant common cover $\longrightarrow$ square-root lifts for phase and transport.

The cover is built from the pair spaces; it is not imposed from outside.

3 Formal II. Spinor one-forms and the meromorphic half-form amplitude vector

3.1 Global phase and transport one-forms

On $\widehat{\mathcal{C}}$, the two local exact-WKB quantities become meromorphic one-forms

$$\omega := \Delta du = \sqrt{y} du, \quad \eta := S du = \frac{\rho}{y} du. \quad (3.1)$$

Here $\Delta = \sqrt{y}$ is understood locally on a chosen phase branch, while η is globally defined once the spinor variable ρ is adjoined.

Remark 3.1. The names *phase* and *transport* are literal. The local WKB exponentials are built from integrals of ω , while the adiabatic mixing is controlled by the transport one-form $\eta = S du$. Thus Σ_Y encodes phase because it remembers the squared gap $y = \Delta^2$, and Σ_R encodes transport because its spinor lift determines S and hence the connection coefficients.

3.2 Half-form packaging

Let $U \subset \mathbb{P}_u^1 \setminus (\text{turning points} \cup \mathfrak{C}_R)$ be simply connected, and let $P_1(u), P_2(u), P_3(u) \in \widehat{\mathcal{C}}$ be the three lifted local sheets above U . On such a domain, the reduced amplitude class defined later in Section 4 has three local scalar carriers R_1, R_2, R_3 , one on each lifted sheet. The normalized adiabatic amplitudes are recovered sheetwise by

$$f_\alpha = \Gamma_\alpha R_\alpha. \quad (3.2)$$

Definition 3.2 (Meromorphic half-form adiabatic amplitude vector). On a regular local sheet of $\widehat{\mathcal{C}}$, the *meromorphic half-form adiabatic amplitude vector* is the object

$$\Psi := \sum_{\alpha=1}^3 f_\alpha e_\alpha \otimes \nu_\alpha = \sum_{\alpha=1}^3 \Gamma_\alpha R_\alpha e_\alpha \otimes \nu_\alpha, \quad \nu_\alpha^2 = du, \quad (3.3)$$

where e_α is a local frame for the corresponding lifted eigenline and ν_α is a local half-form.

The point of (3.3) is not to alter the reduced carriers R_α . It is to normalize them into the adiabatic amplitudes f_α while keeping the whole package on the correct branched geometry. In exact-WKB terms:

$$\text{phase carrier} \leftrightarrow \omega, \quad \text{transport mixing} \leftrightarrow \eta. \quad (3.4)$$

Companion intuition.

The half-form vector is the smallest package that remembers where the amplitudes actually live. If one writes only scalars on the base plane, every continuation across a branch point looks like a confusing mix of conventions, sheet relabelings, and sign choices. The half-form package says: no, the amplitudes live upstairs, on the same space that carries the phase and transport differentials.

In practice this means that the reduced scalar solutions R_α should never again be imagined as just three unrelated functions of u . They are the raw sheet carriers. Multiplying by Γ_α turns them into the normalized adiabatic amplitudes f_α , and those amplitudes are coefficients of a branched geometric object. The integrals of ω tell us which exponential is dominant; the values of η tell us how the local sheets talk to one another.

4 Formal III. From sheetwise evolution to the compact quotient-ring ODE

The starting point is the gauged adiabatic system (1.5). The passage to the reduced carrier is a change of frame: the f_i are coefficients in a moving adiabatic frame, while the functions $R(\lambda_i(u); u)$ are the coefficients of the same solution in the quotient-ring spectral frame. The progression to keep in view is

$$(1.5) \implies \partial_u R(\lambda_i(u); u) = i(E_i - E_x)R(\lambda_i(u); u) \implies \partial_u[R] = i[(m - E_x p)s_u R].$$

The first arrow absorbs the transport connection into the moving spectral frame; the second packages the three scalar sheet equations into one coordinate-free equation in $\mathcal{E}_u$.

A notational point matters here. In the middle equation, $\partial_u R(\lambda_i(u); u)$ means: differentiate the representative $R(\lambda; u)$ with respect to u at fixed λ , then evaluate at $\lambda = \lambda_i(u)$. It is *not* the total derivative of the numerical function $u \mapsto R(\lambda_i(u); u)$. This is why a flat quotient-ring section may still produce nontrivial u -dependence after evaluation on the moving roots.

4.1 Sheetwise scalar evolution

Fix the reference index x . On each spectral sheet, consider the scalar reduced-carrier equation

$$\partial_u R(\lambda_i(u); u) = i(E_i(u) - E_x(u)) R(\lambda_i(u); u), \quad i = 0, 1, 2. \quad (4.1)$$

This is the reduced carrier obtained after removing the moving-frame transport part of (1.5).

Proposition 4.1 (Carrier-to-adiabatic reconstruction). *On each sheet, the normalized adiabatic amplitude is*

$$f_i(u) = \Gamma_i(u) R(\lambda_i(u); u). \quad (4.2)$$

Proof. The Type-1 eigenvector on the i th sheet is

$$\widehat{v}_i = \left(\frac{\gamma_0}{\lambda_i - \epsilon_0}, \frac{\gamma_1}{\lambda_i - \epsilon_1}, \frac{\gamma_2}{\lambda_i - \epsilon_2} \right)^T.$$

Normalizing it multiplies by Γ_i , because $\Gamma_i^{-2} = \Sigma_i^{(2)}$. The reduced carrier $R(\lambda_i; u)$ is the coefficient of the unnormalized vector $\widehat{v}_i$, while f_i is the coefficient of the normalized adiabatic eigenvector. Hence $f_i = \Gamma_i R(\lambda_i; u)$. $\square$

4.2 Interpolation into the quotient ring

For fixed u , there is a unique polynomial representative of degree at most two whose values on the three roots of q_u equal the three reduced sheet carriers in (4.1). Denote its class by

$$[R(u)] \in \mathcal{E}_u := \mathbb{C}[\lambda]/(q_u(\lambda)). \quad (4.3)$$

Let $s_u(\lambda)$ be the inverse of $p(\lambda)$ in $\mathcal{E}_u$:

$$p s_u \equiv 1 \pmod{q_u}. \quad (4.4)$$

Proposition 4.2 (Compact quotient-ring evolution). *The sheetwise system (4.1) is equivalent to the single quotient-ring equation*

$$\partial_u[R] = i[(m - E_x p)s_u R] \quad \text{in } \mathcal{E}_u. \quad (4.5)$$

Equivalently, if

$$h_x(\lambda; u) := \text{red}_{q_u}((m(\lambda) - E_x(u)p(\lambda))s_u(\lambda)), \quad (4.6)$$

then

$$\partial_u[R] = i H_x[R], \quad H_x := M_{h_x}, \quad \nabla_u := \partial_u - i H_x. \quad (4.7)$$

Proof. Evaluate $(m - E_x p)s_u$ at any root $\lambda_i(u)$ of q_u . Since $p(\lambda_i)s_u(\lambda_i) = 1$,

$$((m - E_x p)s_u)(\lambda_i) = \frac{m(\lambda_i) - E_x p(\lambda_i)}{p(\lambda_i)} = E_i - E_x.$$

Therefore the right-hand side of (4.5) agrees with the right-hand side of (4.1) on all three roots. Two classes in $\mathcal{E}_u$ are equal if and only if their representatives agree at all roots of q_u . Hence (4.1) and (4.5) are equivalent. $\square$

4.3 Recovering the adiabatic ODE and lifting to the double cover

Let

$$e_i(\lambda; u) := \prod_{j \neq i} \frac{\lambda - \lambda_j(u)}{\lambda_i(u) - \lambda_j(u)}$$

be the local spectral idempotents. They form the natural moving spectral frame of $\mathcal{E}_u$, satisfy $e_i(\lambda_j(u); u) = \delta_{ij}$, and diagonalize every multiplication operator.

Proposition 4.3 (From the compact equation back to the gauged adiabatic system). *Write the quotient-ring class in the moving spectral frame as*

$$[R(u)] = \sum_{i=0}^2 R_i(u) e_i(u), \quad R_i(u) = R(\lambda_i(u); u), \quad f_i(u) = \Gamma_i(u) R_i(u).$$

Then (4.5) is equivalent to the gauged adiabatic system

$$f'_i(u) = - \sum_{j \neq i} S_{ij}(u) f_j(u) + i(E_i(u) - E_x(u)) f_i(u).$$

Proof. Evaluating (4.5) at the i th root of q_u gives the reduced sheet equation (4.1). Passing from the unnormalized spectral frame to the normalized adiabatic frame multiplies the i th coefficient by Γ_i . The induced connection of that normalized frame is exactly the antisymmetric matrix S_{ij} , so (4.1) together with (4.2) is equivalent to the original gauged adiabatic evolution (1.5). $\square$

Thus (4.5) may be read on four linked levels:

$$\begin{aligned} \partial_u[R] &= iH_x[R] \xrightarrow{\text{evaluate at } \lambda_i} \partial_u R(\lambda_i(u); u) = i(E_i - E_x) R(\lambda_i(u); u) \\ &\xrightarrow{\times \Gamma_i} f_i = \Gamma_i R(\lambda_i(u); u) \xrightarrow{i \mapsto P_\alpha} f_\alpha = \Gamma_\alpha R_\alpha. \end{aligned}$$

The first arrow evaluates the intrinsic bundle equation at a root of q_u . The second restores the normalized adiabatic amplitude. The third lifts the same sheet to $\widehat{\mathcal{C}}$. This is why (4.1), (4.2), and (4.11) all look closely related: they are the same reduced dynamics written before evaluation, after evaluation, and after lifting to the spinor geometry.

The class $[R]$ and the spinor geometry are linked by evaluation on the same sheets. Over a regular domain, evaluation of $[R]$ on the three roots gives the three local reduced carriers

$$R_\alpha(u) = R(\lambda_\alpha(u); u), \quad \alpha = 1, 2, 3, \tag{4.8}$$

the corresponding normalized amplitudes are

$$f_\alpha(u) = \Gamma_\alpha(u) R_\alpha(u), \quad \alpha = 1, 2, 3, \tag{4.9}$$

and the same local sheet carries the corresponding pair data $(\Delta_\alpha, S_\alpha)$, hence the point

$$P_\alpha(u) = (u, r_\alpha(u), y_\alpha(u), \rho_\alpha(u)) \in \widehat{\mathcal{C}}. \quad (4.10)$$

Thus the reduced ODE does not live on a geometry separate from the double cover. Its sheetwise carriers and the reconstructed adiabatic amplitudes are indexed by the very same lifted sheets that carry ω and η .

Corollary 4.4 (Local lift of the reduced ODE). *On a regular local sheet of $\widehat{\mathcal{C}}$, the reduced quotient-ring evolution becomes the scalar equation*

$$\partial_u R_\alpha = i(E_\alpha - E_x) R_\alpha. \quad (4.11)$$

The associated half-form amplitude vector is obtained by first forming $f_\alpha = \Gamma_\alpha R_\alpha$ and then inserting the normalized amplitudes into (3.3).

Special case: vanishing phase data Assume $E_i \equiv 0$ on all sheets, so the phase part of the reduced connection vanishes. Then $h_x = 0$ and (4.5) collapses to

$$\partial_u[R] = 0.$$

Thus the reduced carrier is a flat section of the quotient bundle. For the solution anchored at the diabatic label x , the natural flat section is represented by the u -independent rational function

$$R_x(\lambda) = \frac{\gamma_x}{\lambda - \epsilon_x}. \quad (4.12)$$

Evaluating this flat section on the i th sheet gives

$$R_i(u) = R_x(\lambda_i(u)) = \frac{\gamma_x}{\lambda_i(u) - \epsilon_x}. \quad (4.13)$$

Reconstruction of the normalized adiabatic amplitudes via (4.2) therefore yields

$$f_i(u) = \Gamma_i(u) R_i(u) = \frac{\Gamma_i(u) \gamma_x}{\lambda_i(u) - \epsilon_x}. \quad (4.14)$$

This is the explicit solution to keep in mind. It is exactly the decomposition of the fixed diabatic basis vector $|x\rangle$ in the moving adiabatic basis.

Companion intuition.

The apparent paradox is that $[R]$ is flat while the evaluated quantities $R_i(u)$ and $f_i(u)$ still vary. There is no contradiction. The equation $\partial_u[R] = 0$ differentiates the section at fixed spectral variable λ ; only afterwards does one evaluate it on the moving roots $\lambda_i(u)$.

So the same flat section (4.12) produces the sheetwise carriers (4.13), and multiplying by Γ_i gives the normalized adiabatic amplitudes (4.14). In physical terms, nothing dynamical is left in the reduced carrier: the nontrivial u -dependence is entirely the rotation of the adiabatic basis. That is why the closed form $f_i = \Gamma_i \gamma_x / (\lambda_i - \epsilon_x)$ is the right sanity check.

What kind of function is $R(\lambda; u)$? The notation is easiest to parse if one keeps both variables visible. A representative of the quotient-ring section may always be chosen in polynomial form

$$R(\lambda; u) = c_0(u) + c_1(u)\lambda + c_2(u)\lambda^2, \quad (4.15)$$

with u -dependent coefficients $c_a(u)$. Evaluation on the moving roots then gives the sheet carriers

$$R_i(u) = R(\lambda_i(u); u).$$

Equivalently, one may use the moving idempotent basis and write

$$R(\lambda; u) = \sum_{i=0}^2 R_i(u) e_i(\lambda; u). \quad (4.16)$$

The equation $\partial_u[R] = 0$ means that the class represented by (4.15) is flat at fixed spectral variable λ ; it does *not* mean that the evaluated samples $R(\lambda_i(u); u)$ are constant. In the special flat example one may pick the u -independent rational representative $R_x(\lambda) = \gamma_x / (\lambda - \epsilon_x)$, but in general R is a two-variable carrier field on the spectral line whose coefficients evolve with u and are then sampled on the moving roots.

5 Formal IV. The commuting algebra and the operator realization of the cover

5.1 The commuting triple

The quotient-ring connection $\nabla_u = \partial_u - iH_x$ is only the first multiplication operator. The project also introduces

$$Y := M_\eta, \quad R := M_\tau, \quad (5.1)$$

with symbols

$$\eta(\lambda) = \frac{Q_4(\lambda)L_H(\lambda)^2}{p(\lambda)^2}, \quad \tau(\lambda) = \frac{W_4(\lambda)}{L_H(\lambda)^4}. \quad (5.2)$$

Because H_x , Y , and R are all multiplication operators in the same commutative algebra $\mathcal{E}_u$,

$$[H_x, Y] = [H_x, R] = [Y, R] = 0. \quad (5.3)$$

5.2 Local dictionary between operators and pair data

Proposition 5.1 (Local dictionary). *For $\{i, j, k\} = \{0, 1, 2\}$, the common eigenline labeled by spectator i carries the eigenvalues*

$$H_x \rightsquigarrow E_i - E_x, \quad Y \rightsquigarrow y_i = \Delta_{jk|i}^2, \quad R \rightsquigarrow r_i = \kappa_S \frac{S_{jk}^{-2}}{\Delta_{jk|i}^4}. \quad (5.4)$$

Hence the geometric coordinates (y, r) used to define $\mathcal{C}$ are exactly the spectra of (Y, R) .

Proof. The operator H_x was constructed precisely so that its eigenvalues are the gauged phase differences $E_i - E_x$. The symbols η and τ were chosen so that evaluation on the i th branch of the spectral fiber reproduces the sign-invariant pair quantities in (2.4). Since the operators commute, these values occur simultaneously on one and the same eigenline. $\square$

Corollary 5.2. *The phase curve Σ_Y is the spectral curve of Y , and the transport curve Σ_R is the spectral curve of R . In particular, the formal transport polynomial introduced in Section 2 is not merely a graphing device; it is the actual spectral polynomial of the commuting-family transport operator.*

Companion intuition.

This is the point where the geometry and the algebra finally lock together.

The cover was built first from pair spaces because that is how the data appear in practice. The commuting algebra now explains why that construction was natural: the same coordinates y and r are not arbitrary invariants but the eigenvalues of globally defined multiplication operators.

After this step, “pair sector” stops meaning “one of three charts that we hope will glue” and starts meaning “one common eigenline of a commuting triple.” That shift is exactly what makes the theorem below short.

6 Formal V. The gluing theorem

6.1 Regular set and spectral projectors

Let

$$\chi_R(u, r) := \det(rI - R(u)) \quad (6.1)$$

be the cubic spectral polynomial of R . The transport discriminant locus introduced earlier in (2.18) is equivalently described by

$$\mathfrak{C}_R = \{u \in \mathbb{P}_u^1 : \text{Disc}_r \chi_R(u, r) = 0\}. \quad (6.2)$$

On the regular set

$$U^{\text{reg}} := \mathbb{P}_u^1 \setminus \mathfrak{C}_R, \quad (6.3)$$

the three eigenvalues $r_1(u), r_2(u), r_3(u)$ are distinct. Hence the spectral projectors

$$\Pi_\alpha(u) = \prod_{\beta \neq \alpha} \frac{R(u) - r_\beta(u)I}{r_\alpha(u) - r_\beta(u)}, \quad \alpha = 1, 2, 3, \quad (6.4)$$

are holomorphic.

Definition 6.1 (Global eigenlines). For $\alpha = 1, 2, 3$, define

$$\mathcal{L}_\alpha(u) := \Pi_\alpha(u)\mathcal{E}_u, \quad u \in U^{\text{reg}}. \quad (6.5)$$

6.2 Canonical gluing

Theorem 6.2 (Canonical gluing on the regular set). *On U^{reg} , the three local pair sectors glue canonically into the holomorphic eigenline decomposition*

$$\mathcal{E}_u = \mathcal{L}_1(u) \oplus \mathcal{L}_2(u) \oplus \mathcal{L}_3(u). \quad (6.6)$$

Equivalently, there is no independent pair-sector gluing data on U^{reg} .

Proof. Away from $\mathfrak{C}_R$, the operator R has simple spectrum, so the projectors (6.4) are holomorphic. Because H_x and Y commute with R , they preserve each $\mathcal{L}_\alpha$. By Proposition 5.1, each local pair

sector is exactly one common eigenline of (H_x, Y, R) . Therefore the local pair charts are simply local restrictions of the global splitting (6.6). No additional gluing datum remains. $\square$

Corollary 6.3. *Any ambiguity in sector labeling can occur only at points of $\mathfrak{C}_R$, that is, only where the transport spectral curve ramifies.*

6.3 Local model at a simple transport ramification point

Proposition 6.4 (Square-root branching). *Let $u_* \in \mathfrak{C}_R$ be a simple ramification point of Σ_R . Then, after relabeling,*

$$r_{\pm}(u) = r_* \pm c\sqrt{u - u_*} + O(u - u_*), \quad c \neq 0. \quad (6.7)$$

On the local double cover $t^2 = u - u_$, the two colliding branches become analytic and the corresponding eigenspace splitting extends holomorphically.*

Proof. A simple point of $\mathfrak{C}_R$ is, by definition, a simple zero of the discriminant of the cubic polynomial $\chi_R(u, r)$. The standard local normal form for a simply ramified two-sheet branch gives (6.7). Passing to the coordinate t with $t^2 = u - u_*$ resolves the branch, hence the two eigenvalues and their spectral projectors become analytic in t . $\square$

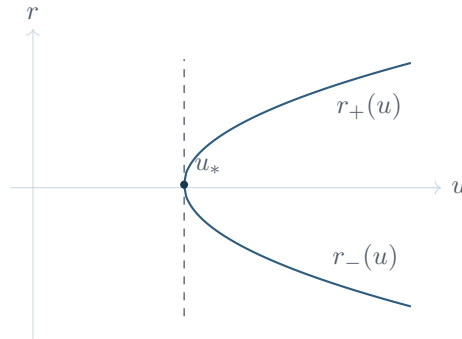


Figure 2: Local form of the transport spectral curve near a simple ramification point. The old sector ambiguity is exactly square-root branching of the transport sheets.

Companion intuition.

The theorem says something stronger than “we found a good gluing rule.” It says that, on the regular set, there was never extra gluing data to begin with.

Before the commuting-algebra picture, it was natural to think that each pair sector came with its own local transport story and that some hidden compatibility matrix had to be supplied to combine them. The theorem shows that the compatibility is already built into the spectral projectors of R .

So the only genuinely nontrivial places are the branch points of Σ_R . There the problem is no longer mysterious chart surgery. It is ordinary spectral ramification.

7 Formal VI. Selector construction and completion of the WKB picture

7.1 Transport ramification plus phase resonance

Let $u_* \in \mathfrak{C}_R$ be a simple ramification point, and let $P_{\pm}(u) \in \widehat{\mathcal{C}}$ be the two lifted sheets whose transport coordinates collide at u_* . Choose a local path $\gamma_{+-}(u)$ on $\widehat{\mathcal{C}}$ connecting these two sheets and define the phase mismatch

$$\mathcal{S}_{+-}(u) := \int_{\gamma_{+-}(u)} \omega. \quad (7.1)$$

Theorem 7.1 (Necessary selector criterion). *Fix a Borel direction θ . A nontrivial selector or new-Stokes event can occur at u_* only if both conditions hold:*

1. $u_* \in \mathfrak{C}_R$, so u_* is a transport ramification point;
2. the phase mismatch is Stokes-real in direction θ :

$$\mathrm{Im}\left(e^{-i\theta} \mathcal{S}_{+-}(u_*)\right) = 0. \quad (7.2)$$

Proof. Condition (1) follows from Corollary 6.3: away from $\mathfrak{C}_R$, the eigenline decomposition is canonical, so there is no sector ambiguity left to activate. Condition (2) is the usual exact-WKB requirement that the two competing exponentials carried by the colliding sheets be phase-aligned in the chosen Borel direction. Since the local carriers are governed by the phase one-form ω , the relevant action mismatch is exactly (7.1). $\square$

7.2 What remains to complete the WKB picture

The theorem above is a reduction, not yet a full solution. To turn the geometric criterion into transition probabilities one still needs four ingredients.

1. **Ordinary sector data.** The ordinary turning points and ordinary Stokes curves coming from the phase curve Σ_Y .
2. **Active selector set.** Among all ramification points of Σ_R , the subset satisfying (7.2).
3. **Local multipliers.** The connection constants attached to each active selector point.
4. **Ordered global assembly.** The correct ordered product of diagonal propagation, ordinary jumps, and selector jumps between the chosen asymptotic regions.

Remark 7.2. The gluing theorem solves the old first question—how the pair sectors fit together. The open work is now concentrated in the second question: which transport ramification points are phase-active and what multiplier does each carry?

Companion intuition.

This is the narrative endpoint a new collaborator should keep in view.

The ordinary pairwise geometry is already explicit. The global gluing is now canonical. So the unresolved content has been compressed into one sharper problem: sift the ramification points of the transport curve through the phase condition (7.2), then compute the local multipliers of the surviving points.

That is why the double cover matters so much. It turns an amorphous three-sector continuation problem into a constrained selector problem on one global geometric stage.

8 Glossary of objects

Table 1: Core objects used in the backbone progression.

Object	Definition	Role
$q_u(\lambda)$	$u p(\lambda) - n(\lambda)$.	Defines the spectral fiber and the quotient algebra $\mathcal{E}_u$.
$\mathcal{E}_u$	$\mathbb{C}[\lambda]/(q_u(\lambda))$.	Rank-three fiber algebra packaging the three sheet values into one class.
$[R]$	Interpolation class in $\mathcal{E}_u$ whose values at the roots of q_u are the three reduced sheet carriers R_i .	Global form of the reduced scalar evolution before adiabatic normalization.
s_u	Inverse of p in $\mathcal{E}_u$, satisfying $p s_u \equiv 1 \pmod{q_u}$.	Removes the denominator in $E = m/p$ inside the quotient algebra.
H_x	Multiplication by $h_x = \text{red}_{q_u}((m - E_x p)s_u)$.	Quotient-ring Hamiltonian for the gauged reduced evolution.
$e_i(\lambda; u)$	Local spectral idempotent $\prod_{j \neq i} (\lambda - \lambda_j) / (\lambda_i - \lambda_j)$.	Moving spectral frame whose evaluations recover the reduced carriers R_i .
R_i	$R(\lambda_i(u); u)$.	Reduced carrier on the i th sheet; the adiabatic amplitude is $f_i = \Gamma_i R_i$.
$\Sigma_{u,\Delta}$	Normalized closure of the graphs $(u, \Delta_i(u))$.	Directly computable phase pair space.

Object	Definition	Role
$\Sigma_{u,S}$	Normalized closure of the graphs $(u, S_i(u))$.	Directly computable transport pair space.
$\Sigma_{\Delta,S}$	Normalized closure of the graphs $(\Delta_i(u), S_i(u))$.	Directly computable mixed pair space.
y	Δ^2 .	Sign-invariant phase coordinate that glues globally.
κ_S	$-\text{Res}_\lambda(p, n) = \gamma_0^2 \gamma_1^2 \gamma_2^2 \prod_{a < b} (\epsilon_a - \epsilon_b)^2$.	Family-invariant normalization constant; positive for real ordered Type-1 data.
r	$\kappa_S S^{-2} / \Delta^4 = W_4 / L_H^4$.	Sign-invariant transport coordinate; more useful than S because it is rational on the common cover before the spinor lift.
χ_R^{loc}	$\prod_i (r - r_i(u))$ on a regular chart; later $\det(rI - R(u))$.	Formal defining polynomial of the transport curve.
$g_\xi(z)$	$(n(z)p(\xi) - n(\xi)p(z)) / (z - \xi) = A_\xi z^2 + B_\xi z + C_\xi$.	Quadratic whose roots are the two sheets complementary to the spectator sheet ξ .
Q_4	$\text{Disc}_z(g_\lambda)$.	Quartic controlling the phase square root and the gap coordinate $y = Q_4 L_H^2 / p^2$.
L_H	$\beta_H + \alpha_H(\epsilon_0 + \epsilon_1 + \epsilon_2 - \lambda)$.	Member-dependent linear factor carrying the chosen commuting-family Hamiltonian into the phase and transport coordinates.
W_4	$np' - n'p = p^2 \Sigma^{(2)}$.	Quartic controlling transport ramification; its zeros are the critical points of the map $u = n/p$.
$\chi_{S^{-2}}$	$\text{Res}_\xi(q_u(\xi), D_{12}(\xi)\sigma - N_{12}(\xi))$, with $\sigma = S^{-2}$.	The compatible (u, S^{-2}) polynomial; cubic in σ and twelfth order in u .
Σ_Y	Curve $\{(u, y) : y^3 - 2Ay^2 + A^2y - D = 0\}$.	Global phase curve for squared pair gaps.
Σ_R	Curve $\{(u, r) : \chi_R(u, r) = 0\}$, equivalently the normalized closure of the graphs $(u, r_i(u))$.	Global transport curve.
$\mathcal{C}$	Fiber product $\Sigma_Y \times_{\mathbb{P}_u^1} \Sigma_R$.	Common phase-transport cover.
ρ	Spinor variable with $\rho^2 = \kappa_S / r$.	Restores the transport square root.
$\widehat{\mathcal{C}}$	Spinor double cover of $\mathcal{C}$.	Natural stage for the transport one-form and the selector construction.
ω	$\Delta du = \sqrt{y} du$.	Phase differential governing action comparison.
η	$S du = \rho du / y$.	Transport differential on the spinor cover.
Ψ	Half-form amplitude vector from Definition 3.2.	Geometry-adapted package of the normalized adiabatic amplitudes $f_\alpha = \Gamma_\alpha R_\alpha$.

Object	Definition	Role
Y	Multiplication operator by $Q_4 L_H^2 / p^2$.	Operator whose spectrum gives $y = \Delta^2$.
R	Multiplication operator by W_4 / L_H^4 .	Operator whose spectrum gives the transport coordinate r .
$\mathfrak{C}_R$	$u(W_4 = 0) = \{u : \text{Disc}_r \chi_R(u, r) = 0\}$.	Ramification locus of the transport spectral cover; image of the critical points of the common-sheet map.
$\mathcal{L}_\alpha$	Eigenspace bundle of R on U^{reg} .	Global version of the old local pair sector.
$\mathcal{S}_{+-}$	$\int_{\gamma_{+-}} \omega$.	Phase mismatch entering the selector criterion.

9 Glossary of conceptual frameworks

Table 2: Conceptual frameworks emphasized in this primer.

Framework	Meaning	Why it matters
Pair-space first viewpoint	Start from the directly computable spaces (u, Δ) , (u, S) , and (Δ, S) .	Matches how the problem actually presents itself before the global geometry is recognized.
Sign-invariant reorganization	Pass from (Δ, S) to $(y, r) = (\Delta^2, \kappa_S S^{-2} / \Delta^4)$.	Explains how the common cover is created from the pair spaces.
Spinor lift	Adjoin $\rho^2 = \kappa_S / r$.	Resolves the transport square root and turns sign ambiguity into deck geometry.
r vs. S distinction	Work first with the invariant r , and recover S only after lifting to $\widehat{\mathcal{C}}$.	Isolates transport branching cleanly; S itself is only a square root of that invariant data.
Half-form packaging	Treat local amplitudes as coefficients of a half-form valued section on the lifted cover.	Prevents branch continuation from being hidden inside coordinate conventions.
Reduced-carrier viewpoint	Move from adiabatic coefficients f_i to reduced sheet carriers $R(\lambda_i; u) = f_i / \Gamma_i$, then to $[R]$.	Makes clear that the off-diagonal S_{ij} term is moving-frame transport, not an extra dynamical potential.
Quotient-ring reduction	Replace three sheetwise scalar equations by one equation in $\mathcal{E}_u$.	Produces the compact evolution (4.5), which may be especially useful for $N = 3$.

Framework	Meaning	Why it matters
Commuting-algebra viewpoint	Regard (H_x, Y, R) as commuting multiplication operators.	Turns pair sectors into simultaneous eigenlines rather than separate local constructions.
Canonical gluing	On U^{reg} , the eigenline decomposition of R is already the global sector decomposition.	Eliminates independent gluing data.
Family-invariant abstract cover	Fix the commuting family once and view different members as different rational coordinates on the same normalized curves.	Explains why the transport curve, the common cover, and the spinor lift are shared across the family even though the visible coordinates y_H and r_H change.
Topological reduction	Identify the common cover with $\mathbb{P}_\lambda^1$, the phase and transport lifts as elliptic double covers, and the fully signed mixed space as a genus-five biquadratic cover.	Tells the reader which cycles, branch points, and periods can possibly matter for selector physics.
Selector geometry	Intersect transport ramification with the Stokes-real phase locus of ω .	Sharpens the open problem to one constrained geometric-exact-WKB criterion.

10 Remaining challenges

Table 3: Open problems after the gluing theorem.

Challenge	Status	Why it matters
Sufficiency of the selector criterion	Open	The present theorem gives a necessary condition. One still needs a proof that every phase-active simple ramification point really contributes a selector event.
Local multiplier calculation	Open	Even after the active points are known, transition amplitudes require the local connection constants attached to those points.

Challenge	Status	Why it matters
Efficient construction of Σ_R	Partly structural, partly computational	The full workflow begins by finding the transport spectral cover and its ramification set.
Numerical continuation on $\widehat{\mathcal{C}}$	Open implementation problem	Practical tests of the selector criterion require reliable tracking of colliding sheets and their phase integrals.
Ordered WKB assembly	Partly understood	Transition probabilities come from the ordered product of ordinary propagation, ordinary jumps, and selector jumps.
Extension beyond $N = 3$	Longer-term	It remains unclear how much of the pair-space to spinor-cover story survives past the cubic case.
Degenerate branch collisions and nongeneric topology	Open but sharply localized	When zeros of Q_4 or W_4 collide, the genera drop and the local selector picture changes; a clean treatment of these nongeneric limits is still needed.

11 Closing summary

This redraft is built around the progression that now appears central for the project:

pair spaces $\longrightarrow$ common cover and spinor lift $\longrightarrow$ half-form package
 $\longrightarrow$ adiabatic ODE (1.5) $\longrightarrow$ sheet carrier $R(\lambda_i; u)$
 $\longrightarrow$ quotient-ring ODE (4.5) $\longrightarrow$ commuting algebra
 $\longrightarrow$ canonical gluing $\longrightarrow$ selector criterion.

The structural message is simple.

The directly computable pair data are real, but they are not yet the right global objects. After reorganizing them through (y, r) and adjoining the transport square root, one obtains a common geometric stage $\widehat{\mathcal{C}}$. The reduced scalar dynamics then package neatly into the quotient-ring equation (4.5); when this intrinsic equation is expanded in the moving adiabatic frame it reproduces the original system (1.5). The commuting operators (H_x, Y, R) then show that the three pair sectors are common eigenlines, so gluing is solved away from the transport discriminant. What remains is the selector problem: determine which ramification points of Σ_R are phase-active and what multipliers they carry.

That is the path this primer is meant to light for a new practitioner.

References

- [1] *Multistate Landau–Zener Solver: Augmented ODE with Gauged Dynamics* (internal summary).
- [2] *A Note on the Sextic Gap Polynomial for $N = 3$ Type-1 Matrices* (internal note).
- [3] *Branch / Spinor Inquiry* (internal discussion note).